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Procedural generation: generating music using ..

Procedural generation using ..

Master's thesis in Cybernetics and Robotics
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Norwegian University of Science and Technology
Faculty of Information Technology and Electrical Engineering
Department of Engineering Cybernetics



ABSTRACT

Write an abstract/summary of your thesis, and state your main findings here.

A summary should be included in both English and any second language, if this is applicable, regardless if the thesis is written in English or in your preferred language. These should be on separate pages, the English version first.

SAMMENDRAG

Write an abstract/summary of your thesis, and state your main findings here.

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PREFACE

Write the preface of your thesis here.

You may include acknowledgements and thanks as part of your preface on this page, or you may add it as a new chapter after the preface.

This thesis is written as the final master project at Norwegian University of Science and Technology (NTNU) during the spring of 2026, supervised by Sverre Hendseth.

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All examples can be found at my github.io page where they can be listened to, numberings are consistent with the thesis for convenience.

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ABBREVIATIONS

List of all abbreviations in alphabetic order:

AI	Artificial Intelligence
LISP	List Processing
LLM	Large Language Models
MIT	Massachusetts Institute of Technology
NTNU	Norwegian University of Science and Technology
OOP	Object Oriented Programming
PG	Procedural Generation
PCG	Procedural Content Generation
WFC	Wave Function Collapse
QM	Quantum Mechanics

INTRODUCTION

For hundreds of years composers have been creating music using external tools like dice to help them generate something new. As early as the 18th century the idea of using randomness to help the composing process has been used. The musical dice game, "Musikalisches Würfelspiel", was used to randomly generate music based on precomposed parts. [1] This is a very basic way of generating music, however by the random nature of the dice it will not be able to create anything coherent and meaningful like a human composer can. Additionally this process only used precomposed parts which were designed to work together. If we want to generate everything from scratch we need a way more sophisticated system. This is where procedural generation comes into the picture. If we base the melody on something random and then use the conventions and "rules" for melodic composition we might be able to generate music that sounds somewhat coherent. Getting external help has not only been used by composers, but also by programmers. In recent years the use of AI in software development has become more and more widespread. As an experiment in this thesis I will compare "old-school" development and AI. I will start with normal development without using AI. Then after I have build up some knowledge with the system I will try to improve upon it using AI tools and evaluate both the output of the code and code itself.

Today the most common way to go about generating music is using AI and Large Language Models (LLM). This enables anyone to create music, however these models are trained on large data sets and imitates what has come before. For this thesis the focus is on the craft of creating music, breaking it down to its very primitive form, only pitches and durations. This will enable the user to have more control over exactly what is being generated. However this requires us to create a system and models that capture the essence of how exactly we are to go about constructing the very basic musical ideas. It will not depend on large data sets and will create a musical ideas that can be processed into a fully fledged musical idea.

If we are able to solve this problem of creating a good structure and model for this kind of generation we will not only have a solution for musical ideas, but this kind of system can solve many kinds of problems.

The problem we want to solve is the problem of creating a coherent system

that will be able to generate music that adheres to certain musical principles. In order to do this we need to consider many things, and system design is a big part of this. When building such a program we need to consider what parts it should consist of as well as how they are supposed to interact. Object Oriented Programming (OOP) is a solid way to approach this system design.

The solution will be a system that is modular, easy to modify and generate new music using newly defined constraints or inputs on structure, motifs etc...

BACKGROUND

This chapter provides the background theory for the project. First we give an overview of procedural generation and some of the different methods used in procedural generation. Then we give an overview of some basic music theory that is necessary to fully grasp the structure of the system. Lastly we cover some about object oriented programming.

2.1 Procedural generation

Procedural Content Generation (PCG), often referred to as just Procedural Generation (PG), is algorithmic generation of content. This content can take many forms including landscape, levels, textures or even 3D models. This leads to smaller file sizes since the program can generate the content using algorithms rather than having everything pre-made and saved.

This section will cover several methods of PG and discuss similarities, differences and most importantly limitations of the methods. Additionally we will take a look at and what we are trying to achieve with the procedural generation in this project and propose a new method that will handle the problem of music generation in a new way.

2.1.1 Wave function collapse

Wave Function Collapse (WFC) is a method used to procedurally generate using either given constraints, or it can also use some input that it analyses and generates the constraints procedurally which are then used to generate something in the same style as the input.

It works by initializing the entire space in a superposition which means that the state is yet to be determined. We call this an *unobserved* state, meaning we don't know what state it is in yet. These unobserved states are not complex numbers like they are in Quantum Mechanics (QM), but the process is similar which is why it gets its name inspired by the field of QM. Randomly assigning parts of the space we are able to collapse¹ some of the *wave function*². In Figure 2.1.1 we

¹collapsing a superposition means that we decide the actual state of the given position in the space

²the wave function is simply the constraints what decide what the state of the object should



Figure 2.1.1: Illustration showing how tiles from the input image is being reused in the procedurally generated output [2].

can see an illustration of how the "tiles" of an input image re-appear in the final image generation.

2.1.2 PG using constraint satisfaction

WIP section

Similar to WFC we have PG using constraint satisfaction. This method uses given constraints for influencing the generation. It usually works by randomly generating something in the given domain, then iterates over the generated data and enforces constraints in order to make it adhere to a model designed by the programmer or end user. After several iteration over the domain we end up with a domain where the constraints are satisfied for all the objects in the domain.

In Figure 2.1.2 you can see how the constraints are set up for this example, and Figure 2.1.3 shows the initial domain with how many different states are left for each cell in the domain as well as the domain after AC-3 propagation. Finally Figure 2.1.4 shows the final domain that satisfies both the local and the global constraints.

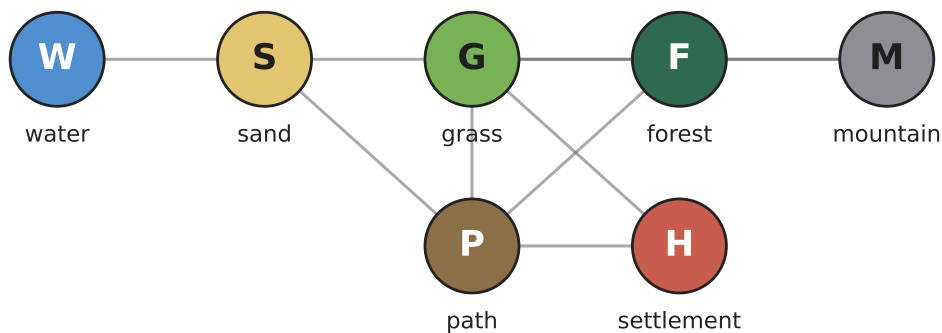
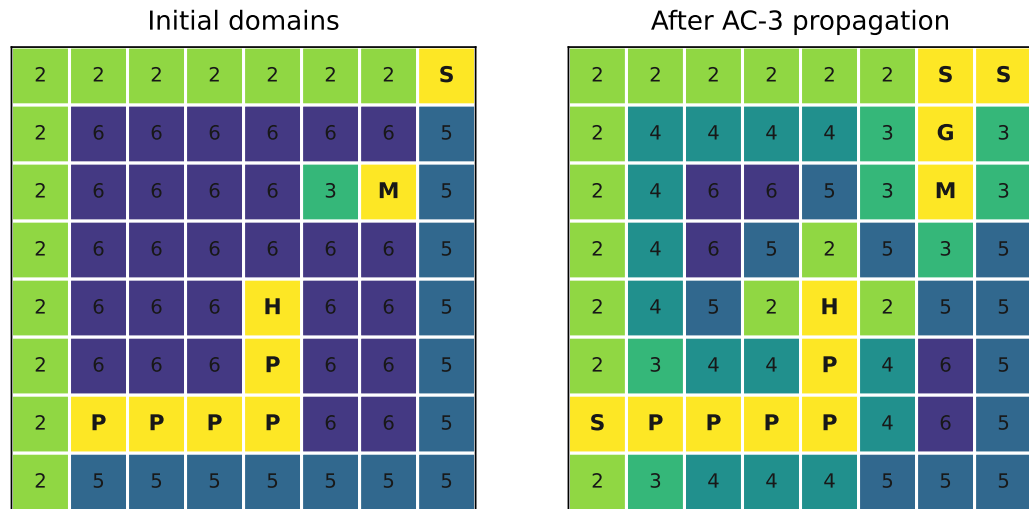


Figure 2.1.2: All different node types and how they are constrained. Each edge represents which node types can be adjacent in the domain.

Constraint propagation reduces the search space



Cell number = remaining domain size; letters mark fixed cells

Figure 2.1.3: Initial domain and domain after AC-3 propagation.

2.1.3 L-systems (Lindenmayer systems)

WIP section

Lindenmayer systems are a method of PG based around the idea of creating rules for the generation which lead to self-similarity and fractal-like forms. This kind of PG is often used in generating organic structures such as trees or other plant-like structures.

If we observe Figure 2.1.5 we can see how two different L-systems can create both very rigid results in the form of the sierpinsky triangle and more organic structures like the fractal plant on the second line.

2.1.4 Binary space partitioning

WIP section

By partitioning the given space in smaller spaces recursively we are able to find a subdivision small enough to generate something that propagates to the higher space with certain characteristics.

2.1.5 Random noise and seeds

WIP section

Perlin noise is a very common way to generate structures like worlds etc... It is very good at creating smooth transitions from one state to another, like height maps.

Generated map satisfying local and global constraints

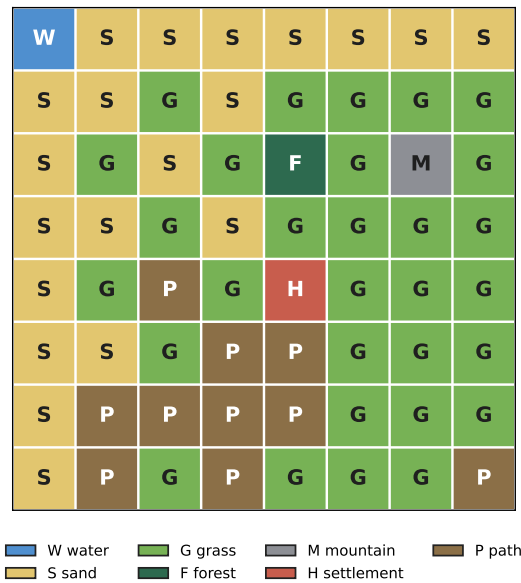


Figure 2.1.4: Final generated map satisfying all constraints.

2.1.6 Random walks

Random walk procedural generation method that can be used to organic looking structures like maps and textures. It can also be used to describe natural phenomena like bacterial movement. It has been applied in financial economics and computer science

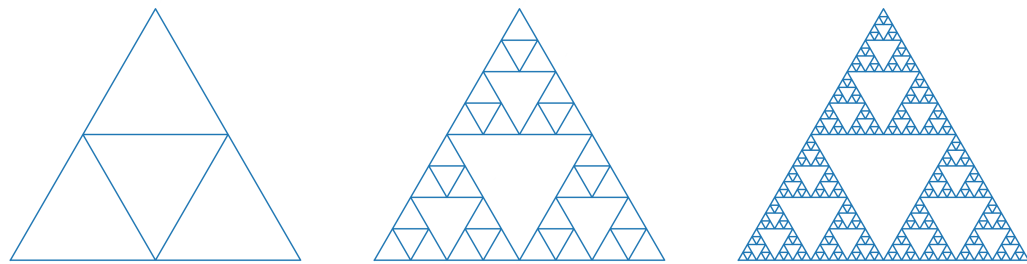
The random walk works by having an entity walk around randomly. In some cases the random walk is also called the *drunkard's walk*. Then the path is traced, either to create a circumference of an area or the traversed spaces become the area itself. Oftentimes the drunk walker has a set amount of time or steps it can walk before it is terminated, then a new walker is placed at a random location, this helps to smooth out the generated area.

The same technique can be used to create textures as well, in Figure 2.1.7 we can see how changing the colours and directional bias we can generate textures that imitate real life. In the tree and sand example the directional bias is vertical for the tree texture and diagonal for the sand texture. For the map texture it is uniform directional bias, but colouring to indicate water, land and mountains. The least traveled to square get more blue, and the more frequent it goes to green and brown/white to indicate mountains.

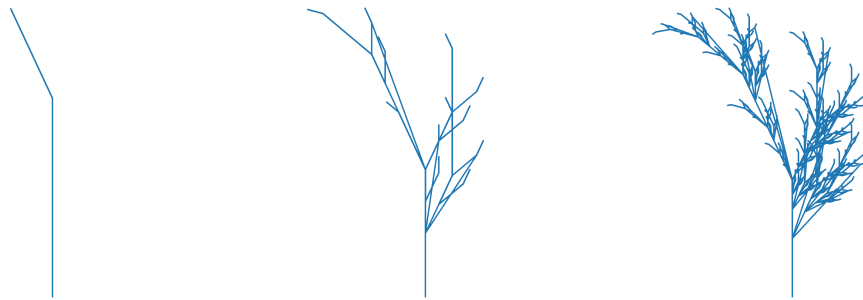
2.1.7 Discussion

WIP section

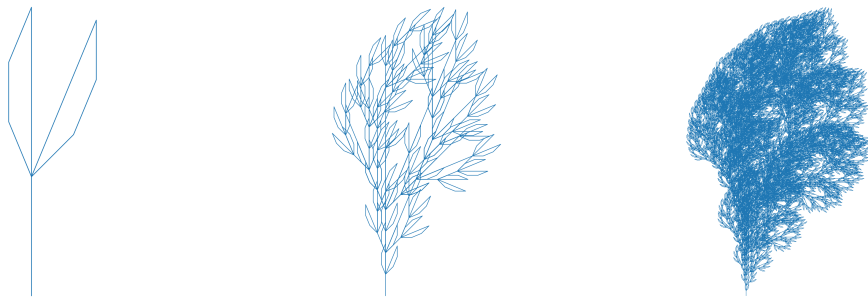
Now that we have taken a look into a few methods of PG we observe that there are many similarities between the methods. Common for all forms of PG is that they utilize some form of randomness. The randomness in a core part of procedural generation, if there was no randomness we would generate the same output every time and that would be the same as just creating the content yourself.



(a) sierpinksi triangle



(b) fractal plant



(c) fractal bush

Figure 2.1.5: Demonstration of the sierpinksi triangle, a fractal plant and bush after one, three and five iterations repsectively.

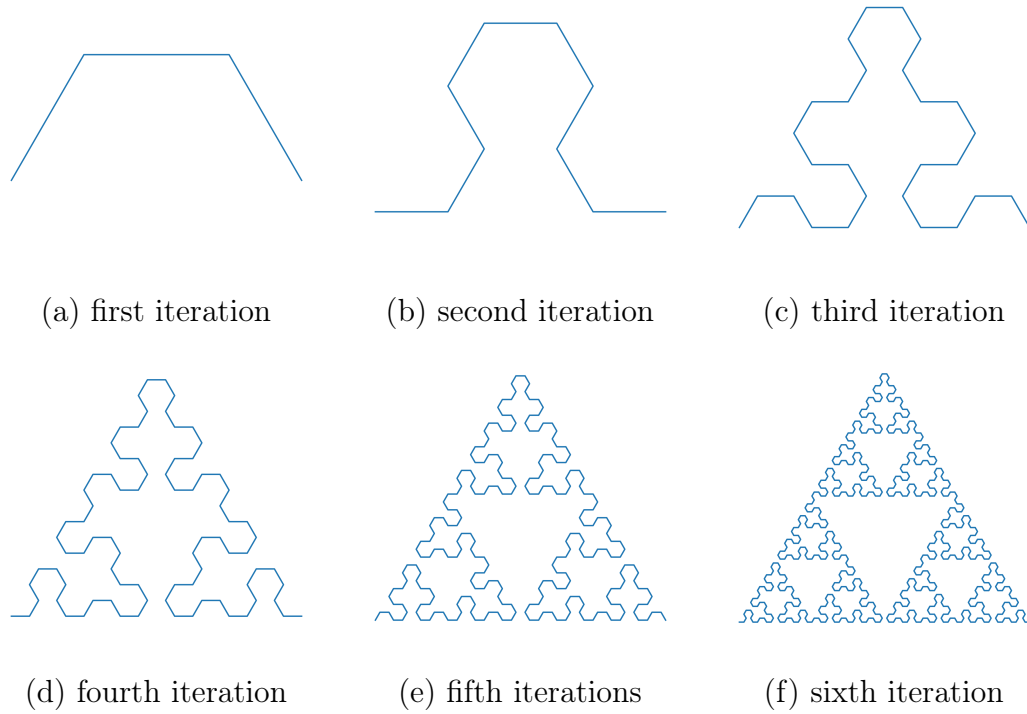


Figure 2.1.6: Approximation of the sierpinski triangle using the sierpinski arrowhead curve.

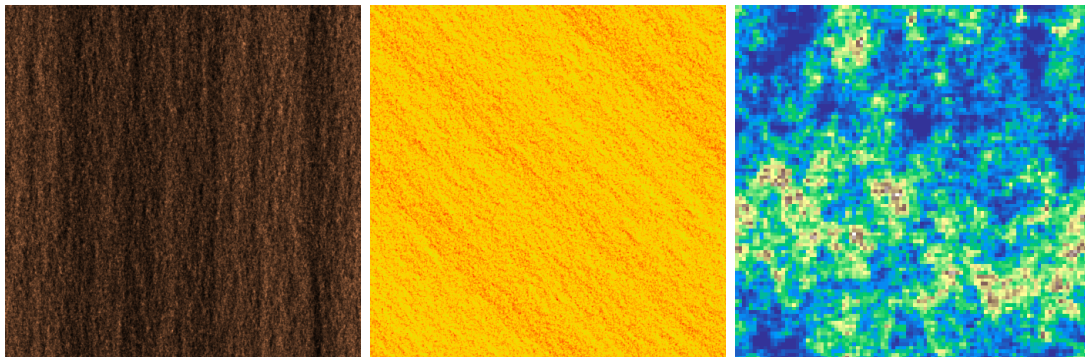


Figure 2.1.7: Three different textures generated using random walks.

Table 2.2.1: Table with common note values, their names and their corresponding duration.

Symbol	Name	Rest symbol	Duration
	Whole note		4 beats
	Half note		2 beats
	Quarter note		1 beat
	Eighth note		1/2 beat
	Sixteenth note		1/4 beat
	Thirtysecond note		1/8 beat

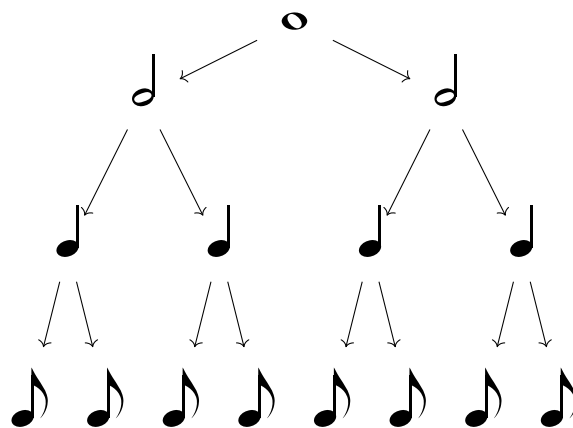


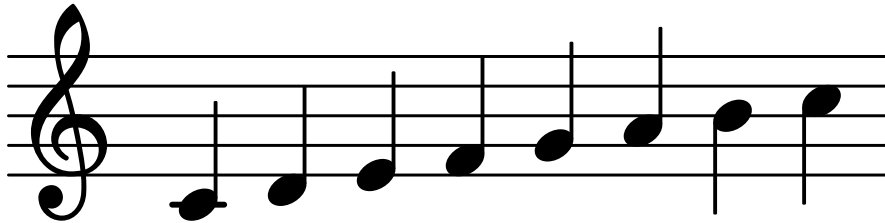
Figure 2.2.2: Tree structure showing how you can divide the whole note into the smaller note values. Here each level in the tree takes the same amount of time, meaning the whole note at the top takes as long as the eight eighth-notes at the bottom.

notes. You can see the basic c-major scale in Figure 2.2.1, it is comprised of the notes c d e f g a b c. The tones of the scale can be named according to their function within the scale. We often label the notes $\hat{1}$, $\hat{2}$, $\hat{3}$, $\hat{4}$, $\hat{5}$, $\hat{6}$, $\hat{7}$, and then starting back at $\hat{1}$. The use of the carets are called the Nashville Number System, this is used to differentiate the scale degrees from the chord tones which will be discussed in Section 2.2.7 When we consider chord structures we use the numbers but usually count above from 8 to signify harmonic functions, but when considering only scales and melodies we usually only use 1 through 7

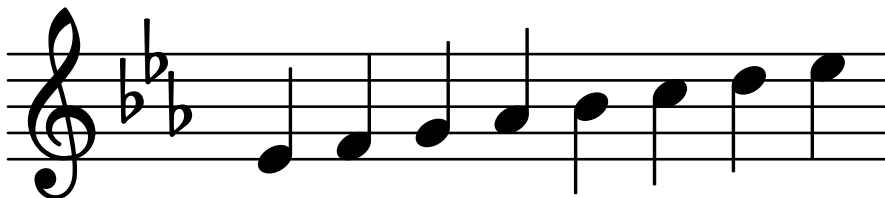
The most important function is that of the scale degree 1, this is the *tonic*, which is the 'home' of the key. This is the most stable function there is, it feels resolved and does not have any tension urging it to move anywhere.

Next up we have the dominant function, there are two notes degrees which fall into this category, that is the 5th and 7th scale degree. These notes have tension. They want to move, especially the 7th scale degree, it has a great pull toward the tonic. You can hear this in the example under Example 2.2.1. The penultimate tone is the 7th scale degree and it leads to the tonic.

To lead us to the tonic functions we have what are called the pre-dominant function. This simply means 'before dominant', and their function is to set up the tension to lead us to the dominant degrees which in turn leads us back to the tonic.



Example 2.2.1: The C-major scale written in the treble clef.



Example 2.2.2: The Eb-major scale written in the treble clef.

2.2.4 Keys

The concept of a musical key is simply shifting around where the tonic note is. For instance in Example 2.2.1 you can see the major scale in C. And in Example 2.2.2 you can see the major scale in Eb. In today's tuning system there are no differences between the keys, meaning that they sound the same, just higher and lower. This was not the case earlier in history, where the temperament (how the tones were tuned in relation to each other) of the keys differed. Today it just sounds higher or lower depending on the key.

2.2.5 Melodies

Melodies is the horizontal component of music, meaning we look at it as a time-series. We have a given note at a given length followed by another note or rest of a given length etc... This means that you can look at a melody as a series of intervals as well. If we do this on known melodies we will observe that they are mainly composed of steps and skips. This means that the distance between one note and the next is one or two tones. Large leaps are usually leaping to important tones like the tonic or the root of the current chord.

2.2.6 Intervals

Before discussing harmony we need to consider the different types of distances between notes. The musical distance between two notes are called an *interval*.

Table 2.2.2: Table containing all musical intervals up to an octave, the number of semitones and their abbreviated form.

Interval name	Semitones	Abbreviation
Unison	0	P1
Minor second	1	P1
Major second	2	m2
Minor third	3	m3
Major third	4	M3
Perfect fourth	5	P4
Tritone ³	6	TT or aug4/dim5
Perfect fifth	7	P5
Minor sixth	8	m6
Major sixth	9	M6
Minor seventh	10	m7
Major seventh	11	M7
Octave	12	P8

Intervals can be identified by the number of semitones between the notes. Additionally we have special names for all the intervals, these names can be seen in Table 2.2.2. The list also provides the number of semitones between the notes.

In Figure 2.2.3 we can see how all the normal intervals look on a staff. Naming the interval on a general basis means looks at the distance, this means that a second is two consecutive notes and a third has "one space" between the notes, this is regardless whether it is minor or major a third looks the same distance wise regardless of quality. You can see this in Example 2.2.3, here we observe that the distance between the notes in a minor third and a major third is the same, the only difference is that the minor third is one semitone smaller, but that is shown by the $\flat$ -symbol which means the note has been lowered by a semitone. Consequently for the tritone we see there are two ways to spell the same interval, one is a fourth that has been raised and the other is a fifth that has been lowered. These two are *enharmonically equivalent* which just means that they are the same pitches, they are just spelled differently. For the augmented fourth(the first spelling of the tritone) we see the $\sharp$ -symbol used, which simply means the note is raised by a semitone.



Example 2.2.3: All the musical intervals from the unison to the octave including their variants.

³can also be called diminished fifth or augmented fourth, where diminished simply means lowered by semitone and augmented means raised by one semitone

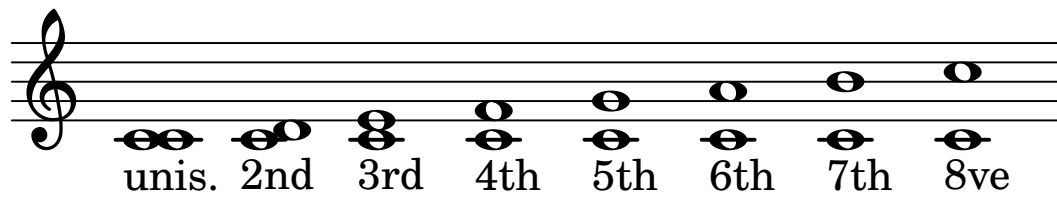


Figure 2.2.3: All normal intervals from the note C₄ devoid their variants in quality.

2.2.7 Harmony

Harmony is the vertical component of the music, harmony is what happens when two or more notes happen at the same time. This is what chords are build around. When considering the harmonic progression in tonal music we can consider all the standard chords in the major key. They are labeled by numbers using the roman numerals. Upper case means the chord in major and lower case means minor. You can see the difference between major and minor chords in figure [make a figure for this], and listen to the difference at the link provided in the caption.

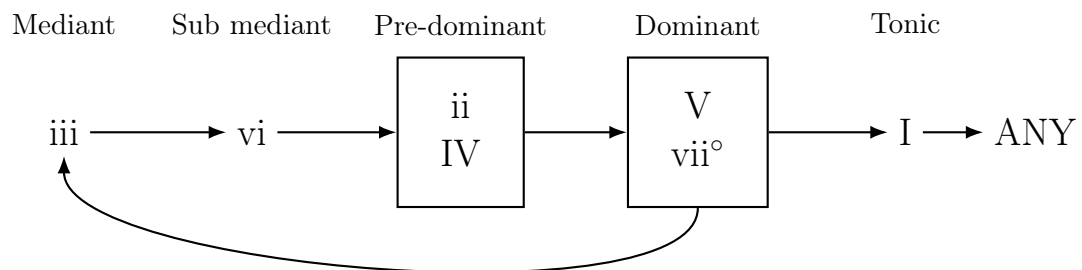
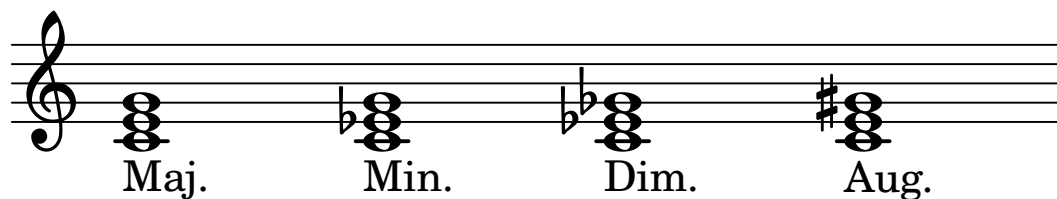


Figure 2.2.4: Harmonic progression in tonal music. The arrows indicate the typical flow of harmonic functions.



Example 2.2.4: The four main triads(chords with three notes), major, minor, diminished and augmented.

2.2.8 Time signatures

The time signature explains what is in a bar of music. A bar is the space between the vertical lines on the staff. A 4/4, sometimes written as **C**, time means that there are four quarter notes per bar and consequently 3/4 time means that there are three quarter notes per bar.



Figure 2.2.5: Demonstration of a few common time signatures.

2.2.9 Musical form

Now that we have an idea of the musical structures on the melodic and harmonic layer we will take a look at musical forms. This is a very important idea when it comes to constructing melodies. A melody is not an infinitely wandering idea of music. A good melody follows a form. Often this can be viewed as repetitions and variations of a given musical idea. Two very common forms of musical form are the *sentence form* and the *period form*.

In Figure 2.2.6 we can see an example of the musical sentence. It starts with a 2-bar opening phrase, labeled as the *basic idea*, then we have a repetition of the same idea only adapted for a new chord. Then the following 2 bars are a fragmentation of the idea. Then for the last two bars we have a conclusion, this ties the idea together. The cadence at the end can be considered the end of a sentence. It concludes the idea before it moves on to something else.

Figure 2.2.6: A classical example of the musical sentence form. These are the opening bars of Bethovens piano sonata in F-minor. [3]

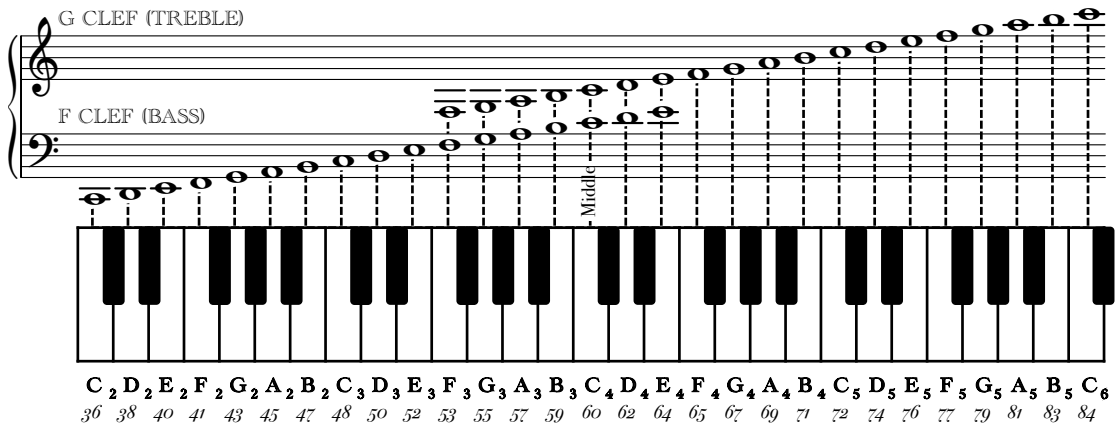


Figure 2.2.7: Overview of note placement on the staff and piano, additionally the corresponding midi numbers are written below the note names. Notice how the number increases by two when there is a black key between consecutive notes.

2.3 Object oriented programming

OOP had its inception in the sixties with the *artificial intelligence* group at Massachusetts Institute of Technology (MIT). Here the term "object" was used in the language family List Processing (LISP) to refer to LISP atoms⁴ with certain attributes or properties. [4]

2.3.1 Features of OOP

Some of the useful features of OOP are that we can achieve using OOP is ...

⁴a LISP atom is an indivisible entity in the language. LISP consist of two data types, atoms and lists. And a list can only contain atoms or other lists

2.4 Iterative development

A major problem in the design of a computer system is that many aspects of system behaviour and performance are not discovered until the system has been built...[5]

F.W. Zurcher

B. Randell

This project utilizes a form of development called iterative design. This means that I do the project by creating a design, taking notes on what needs to be improved, then starting again with the new knowledge gained in the previous iteration.

This sounds like more work, but in reality it leads to an effective workflow. It allows for fast prototyping since we can start programming something straight away, then when the project halts due to the system that has been created is incompatible with the ideas for improving it we take notes on what worked, what is missing, and use this to create a specification for the next iteration.

By doing this we maintain active development which doesn't stagnate and we can always create a new iteration when development slows down. After 3-4 iterations we should be able to land on a system that works in a way that we are happy with.

2.5 AI in software development

In addition to doing this iterative development we will attempt to develop using artificial intelligence as a tool for assisting the process. However the first two iterations will be without any AI. This is to make sure I understand the domain and make my own decisions of how I want the program to work. By doing this work to begin with I work up knowledge in the field and will be better equipped to understand what the AI produces and make sure that it does what I want it to. Without this knowledge I will not be able to efficiently grasp the AI output and will spend a significant time learning what the AI has done.

I will discuss the use of AI and whether it is a tool worth using or if it simply ends up creating more work since we need to manually fix bugs or do prompt engineering

SOFTWARE DESIGN

Include the complete description of the methods used in your research here.

3.1 Object oriented programming

This project will use the widely known process of object oriented programming. In simple terms this simply means that all the parts of the code are structured as objects, meaning everything is represented as objects with given properties. This means that for instance a 'melody object' could consist of all the notes and durations it consists of, the key it is written in and any metadata explaining what the melody represents. Meaning if it is a verse or chorus etc...

In order to design an object oriented program the design process is very important. Creating good models for the objects are imperative to make it work. If the level of abstraction is too high, the semantic gap between the real-world concepts they represent and the actual implementation in the code will be too large and the model will be un-intuitive and hard to both use and also understand.

IMPLEMENTATION

The implementation of this program started by just writing something that generated something. This was done to not get stuck in the planning phase. You could consider this a contradiction to the part in software design which states that good planning is imperative. However, trying out some code and figuring out what works, what doesn't and what needs to be implemented in the models is very helpful. Therefore the first iteration was a test to get some systems working and finding out what it necessary to make it work and what is not important.

4.1 The iteration process

After the first iteration lots of issues were observed and were put into the spec for the second iteration which improved upon the model in various ways. Some important features were restricting the range so that an actual instrument can play it, be that an instrument or the human voice.

DISCUSSION

5.1 Iterative development

5.2 AI development

5.3 The system structure

5.4 Modules? Different module discussions.

5.5 The generated music

5.6 Future work

5.6.1 Improving/Expanding the system

CONCLUSIONS

This project has shown the development of a procedural music generator. The process utilized iterative development as well as AI assistance with developing the system. This resulted in a modular system which encapsulates musical ideas in an effective manner which is easy to understand and modify.

The use of AI proved to be a very useful tool to speed up development both in terms of troubleshooting or debugging as well as helping find solutions to the problems you are working on. The issue with AI is that it removes the creative element of programming and simply regurgitates from the ocean of information it has access to. This means that it solves the problems in similar ways they have been done before, this hinders true innovation and creating something completely new.

On the other end we have the problem that AI is very good at generating content, meaning by prompting AI to solve a problem for us it generates a lot of code, this means that in addition to formulating the problem in such a way that AI can solve it we also need to go over the code and check that it actually does what we intended it to as well as it being maintainable and easy to read. In order to do this part effectively we need to understand not only the language we are developing in, but we also need knowledge of the existing code base. For this problem it means that we need to know not only how the language works, but also the language of music theory. If we do not understand this then we would not be able to make any meaningful changes to the code by ourselves. Let alone if we try to prompt engineer without knowing anything about music.

There seems to be a sort of silver lining in the use of AI when developing software. Using it to develop entire systems will lead to the developer not understanding how the code works or having to spend a substantial amount of time reading through the AI generated code to figure out how it works. Using AI in smaller perspectives can yield better results. It is very good at saving you time by completing trivial tasks that just takes time or debugging small functions etc. that you could have spend minutes or hours tinkering with to solve yourself. This keeps the code your own, but it makes you faster by helping you with the small tasks.

Overall the system generates passable music, there is no defined way to actually grade music objectively, but at least we know that the output follows a few

principles associated with good melodic and harmonic structure.

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APPENDICES

A - GITHUB REPOSITORY

All code and latex-files used in this document are included in the Github repository linked below. Further explanations are given in the readme-file.

Github repository link

- <https://github.com/lima98/TTK4900>

B - AUDIO EXAMPLES

Audio-files for the examples in the thesis and code documentation can be found at my github.io page.

Audio example archive

- <https://lima98.github.io/TTK4900>